

■ Field Watch

Solar-ready Soil Health Monitoring & Farmer Dashboard

Project README — Tech-a-Thon 5.0, Problem Statement 1 (Core Track)

This document explains the complete system: the ESP32 sensor node, how it talks to the cloud database, and every feature of the farmer-facing website — including the admin content panel.

Component	Role
ESP32 Dev Module	Reads sensors, connects to WiFi, sends data to the cloud
Soil moisture sensor	Measures soil water content (0–100%)
DHT11 sensor	Measures ambient temperature and humidity
I2C LCD (16x2)	On-node live display, no laptop needed for demo
Supabase	Cloud database — stores readings, settings, and site content
Website (Netlify)	Farmer dashboard — live data, history, weather, admin panel

1. System Architecture

Data flows in one direction from the field to the farmer, and settings flow back the other way:

ESP32 (WiFi) → HTTP POST → **Supabase** (cloud database) ← reads ← **Website** (Netlify)
Farmer sets reading frequency / threshold / location on the website → saved to Supabase → ESP32 checks and applies it on its next cycle.

This means the ESP32 never talks to the website directly — both sides only talk to Supabase, which keeps the system simple and lets multiple devices (phone, laptop, tablet) see the same live data at once.

2. Hardware & Wiring

2.1 Power

The ESP32 needs 3.3V internally but accepts 5–12V on its **VIN** pin through the onboard regulator. A 6V source (e.g. 4×AA battery pack) can be connected directly to VIN + GND. Never connect an external voltage source to the **3V3** pin — that pin is a regulated output for powering sensors, not an input.

2.2 Pin connections

Component	Pin	ESP32 Pin
Soil moisture sensor	VCC	3V3
Soil moisture sensor	AOUT	GPIO 34
Soil moisture sensor	GND	GND
DHT11	VCC	3V3
DHT11	DATA	GPIO 4
DHT11	GND	GND
I2C LCD	VCC	3V3
I2C LCD	SDA	GPIO 21
I2C LCD	SCL	GPIO 22
I2C LCD	GND	GND
External 6V supply	+	VIN
External 6V supply	–	GND

All sensor VCC and GND lines share the ESP32's own 3V3 and GND pins — use a breadboard power rail rather than stacking multiple wires into one header pin.

3. ESP32 Firmware Setup

3.1 Required libraries (Arduino IDE → Library Manager)

- **DHT sensor library** by Adafruit (also installs Adafruit Unified Sensor)
- **LiquidCrystal I2C** by Frank de Brabander
- **ArduinoJson** by Benoit Blanchon

3.2 What the firmware does each cycle

- Connects to WiFi (2.4GHz only — ESP32 cannot join 5GHz networks)
- Reads soil moisture, temperature, and humidity
- Displays the readings on the I2C LCD
- Sends the reading to Supabase via an HTTP POST request
- Fetches the farmer's current interval and threshold settings from Supabase
- Waits for the configured interval, then repeats

3.3 Values to configure before uploading

Open the sketch and set your own WiFi credentials and Supabase project details near the top of the file:

```
const char* WIFI_SSID = "your_wifi_name";
const char* WIFI_PASSWORD = "your_wifi_password";
const char* SUPABASE_URL = "https://your-project.supabase.co";
const char* SUPABASE_ANON_KEY = "your-anon-public-key";
```

The **anon public key** is safe to include in code — it is designed to be public-facing. Never use the separate `service_role` key anywhere in device or website code.

4. Supabase Setup (Cloud Database)

Supabase provides a free hosted Postgres database with an automatic REST API — the ESP32 and the website both read and write to it directly, with no custom backend server required.

4.1 Tables used

Table	Purpose
readings	Every soil / temperature / humidity reading, timestamped
device_config	Farmer-set reading interval, dry-soil threshold, and saved location
site_content	Editable website content — hero text, about text, gallery, videos, contact info

Row Level Security (RLS) is enabled on all three tables with a permissive policy so the ESP32 and website can read/write using only the public anon key — appropriate for a hackathon demo; a production deployment would tighten these policies.

5. The Farmer Website

A single-page site (deployed free on Netlify) that farmers open on a phone or laptop. It refreshes automatically every 15–20 seconds — no manual reload needed.

5.1 Live monitoring

- Live soil moisture, temperature, and humidity cards, with time-since-last-reading
- 7-day trend chart (soil moisture vs. temperature)
- Recent readings table, with CSV and PDF export buttons
- Dry-soil alert banner when moisture drops below the farmer's chosen threshold

5.2 Farmer controls

- **Reading frequency** — choose from 1 minute up to 6 hours; the ESP32 picks it up automatically
- **Dry soil threshold** — sets when the alert banner activates
- **Field location** — type a village/town name or tap "Use my GPS"

5.3 Location-aware features

- **5-day weather outlook** — live forecast for the saved location (Open-Meteo, free, no key required)
- **Crop suggestions** — a rule-of-thumb match based on temperature, rainfall, and humidity at that location. This is clearly labeled as indicative, not a substitute for local agricultural extension advice
- **Irrigation advisor** — combines the current soil reading with the rain forecast to recommend "Irrigate now", "Hold off — rain expected", or "No action needed"

5.4 Multilingual interface

The interface can be switched between English, Hindi, Tamil, and Telugu from the header dropdown. The chosen language is remembered on that device for future visits.

5.5 Admin panel

An "Admin" link in the footer opens a password-gated panel (password set by the project owner). Once unlocked, the admin can edit:

- Hero title, subtitle, and logo image
- About-section text and any number of extra subtopics
- A photo gallery (image URLs)
- Embedded videos (YouTube or direct video links)
- Contact phone, email, and address

All admin changes save directly to the shared Supabase database, so they appear on every visitor's device automatically within about 20 seconds — no redeploy needed.

Note: the admin password check happens in the browser, which is appropriate for a hackathon demo but is not equivalent to secure server-side authentication. A production version would use proper account-based login (e.g. Supabase Auth).

5.6 Report export

- **CSV export** — raw readings for spreadsheet analysis
- **PDF export** — a formatted summary report generated in-browser

6. Deployment Checklist

- Create a Supabase project and run the SQL setup scripts for all three tables
- Copy the Project URL and anon public key into both the ESP32 sketch and the website file
- Upload the sketch to the ESP32 and confirm WiFi connects and readings POST successfully (Serial Monitor)
- Open the website file locally in a browser and confirm live data appears
- Deploy the website by dragging the HTML file into Netlify's deploy area
- Set the field location and reading frequency from the live site
- Log into the admin panel once to add project photos, videos, and contact details

7. Future Scope

- Solar panel + charge controller (TP4056) for fully off-grid, self-sustaining operation
- LoRa long-range transmission for sites without WiFi coverage
- SMS/WhatsApp alerts for farmers who do not check the dashboard daily
- Multi-node support for monitoring several fields from one dashboard
- On-device buzzer/LED alert wired directly to the dry-soil threshold